

Taiwanese Domestic Supply Chains: 2013–2022*

Teddy C.W. Chen[†] Joonkyo Hong[‡] Toshiaki Komatsu[§]
Jin-Tan Liu[¶] Jen-Kuan Wang^{||}

May 22, 2026

Abstract

This paper documents the structure of Taiwan’s domestic production network from 2013 to 2022, using administrative value-added tax and business-to-business (B2B) transaction records spanning nearly all nonfinancial private-sector firms. We describe the institutional features of Taiwan’s VAT system and examine the internal consistency of the data. The network has grown steadily over the sample period and is highly skewed: a small number of large firms—concentrated in wholesale, retail, and high-tech manufacturing—account for a disproportionate share of domestic linkages and transaction values. By integrating customs records, we decompose each firm’s direct and indirect exposure to foreign markets. Although fewer than 15% of firms are direct importers or exporters, most firms source inputs from importers or sell to exporters within four degrees of separation, making them susceptible to foreign shocks propagated through wholesalers, retailers, and semiconductor producers.

*The views expressed in this paper are those of the authors and do not necessarily reflect the views of any institution with which the authors are affiliated. We are grateful to Kim-Ning Wong and Yi-Chin Chang for helping us understand tax regulations in Taiwan. Hong acknowledges support from NSTC grant 114-2410-H-002-285-MY2. Komatsu acknowledges support from NSTC grant 114-2410-H-002-287-MY2 and the Yushan Fellow Program by the Ministry of Education. All remaining errors are our own.

[†]Washington University in St. Louis. cchia-wei@wustl.edu.

[‡]National Taiwan University. jkhong@ntu.edu.tw.

[§]National Taiwan University. tkomatsu@ntu.edu.tw.

[¶]National Taiwan University and NBER. liujt@ntu.edu.tw.

^{||}Pennsylvania State University. jkwang@psu.edu.

1 Introduction

As of 2019, Taiwan accounted for more than 60 percent of global semiconductor foundry revenue and 92 percent of the world’s most advanced chip manufacturing capacity (Varas et al., 2021). Its role in global value chains, along with rising geopolitical tensions in the region, has drawn considerable policy and academic attention. Existing empirical research on Taiwanese firms has focused primarily on firm heterogeneity, pro-competitive gains from trade, and trade dynamics (Aw and Lee, 2008; Aw et al., 2011; Edmond et al., 2015; Kao and Liu, 2022; Kao et al., 2024). While these studies rely on variations among a small set of large, productive firms that account for the majority of exporting and importing activities, the broader ecosystem of domestic supply chains also plays a crucial role in shaping Taiwan’s industrial development. Mapping Taiwan’s domestic supply chains is therefore essential for understanding how trade, geopolitical, and industrial policy shocks propagate through the economy. Nonetheless, basic facts remain largely undocumented, in part because comprehensive firm-to-firm transaction data have until recently been unavailable.

In this paper, we document the structure of Taiwan’s domestic production network from 2013 to 2022 using newly accessible administrative data on business-to-business (B2B) transactions, compiled from value-added tax records, firm-to-firm sales records, customs records, and individual income tax records maintained by Taiwan’s Fiscal Information Agency. The resulting dataset is an annual panel of bilateral relationships among nonfinancial private-sector firms from 2013 to 2022. It allows us not only to observe the value of each transaction, but also to link each buyer-supplier relationship to the characteristics of both firms, including domestic sales, export and import values, and labor costs. Beyond documenting empirical regularities, we also describe the institutional features of Taiwan’s VAT and B2B reporting systems, examine and validate the internal consistency and quality of the data, and provide details on the construction of a cleaned sample suitable for production network analysis.¹

The empirical patterns we document focus on four aspects of the Taiwanese domestic production network. First, we examine the trend in Taiwanese firms’ reliance on B2B transactions, on both the sales and input-sourcing sides. Second, we study the skewness in the cross-sectional distribution of buyer-supplier linkages across firms and sectors. Third, we explore the concentration of purchases within buyers and of sales within sellers across their trading partners. Fourth, we assess firms’ indirect exposure to foreign markets by explicitly

¹This joins similar datasets from other countries, such as Belgium (Dhyne et al., 2025), Chile (Arkolakis et al., 2025), Costa Rica (Alfaro-Urena et al., 2022; Alfaro-Ureña et al., 2026), Italy (Borin et al., 2025), and Turkey (Demir et al., 2024) that together have reshaped the study of production networks (for a cross-country survey, see Bacilieri et al., 2026). Taiwan is, to the best of our knowledge, the first advanced economy in East Asia with comprehensive administrative value-added tax records on firm-to-firm transactions of this kind.

accounting for transactions with direct importers and exporters.

Each pattern carries potentially broader implications. First, we document a steady expansion of the domestic production network in Taiwan, suggesting its growing importance in shaping firm performance and aggregate production. From 2013 to 2022, domestic B2B transactions increased steadily in both the number of buyer-seller links and the total value of firm-to-firm sales. The number of buyer-seller pairs rose from about 33 million to more than 45 million. The value of domestic firm-to-firm sales grew from roughly 17 trillion NTD to nearly 24 trillion NTD, compared with nominal GDP growth from 15 to 23 trillion NTD over the same period.

Second, consistent with findings from other countries (see, e.g., [Bernard et al., 2022](#)), we show that the distribution of buyer-supplier linkages is highly skewed, with a small set of firms connected to disproportionately many buyers and suppliers. In our 2017 cross-section of buyer-seller linkages, the average Taiwanese firm has about 70 buyers and 70 suppliers, whereas a 95th-percentile firm has 199 buyers and 219 suppliers. At the within-firm level, both purchases and sales are highly concentrated among the top five trading partners. On average, a firm’s top five domestic trading partners account for roughly 80 percent of its domestic B2B purchases and sales.

Third, we show that a few key sectors play central roles in Taiwan’s domestic production network. High-tech manufacturing firms, such as producers of electronic parts, account for only 0.8 percent of all Taiwanese firms but have 172 suppliers on average. This is more than twice the national average of about 70 suppliers, indicating that these firms are major buyers in the domestic economy. By contrast, high-tech knowledge-intensive service firms, such as telecommunications providers, account for 1.3 percent of all firms, while the average firm in this sector has 201 buyers, more than four times the national average, suggesting that these firms are key suppliers. Taken together with the skewed distribution of firm linkages, these patterns imply that shocks to a small number of leading firms in key sectors may propagate widely through domestic production linkages.

Fourth, we find that although fewer than 15 percent of firms engage in direct exporting or importing, the vast majority are indirectly exposed through purchases from direct importers and sales to direct exporters within four transaction links. In particular, excluding the direct trade of wholesalers and retailers reduces the sales-weighted average total import share from 53 to 31 percent, while excluding semiconductor exports reduces the sales-weighted average total export share from 46 to 36 percent. These results highlight the central role of domestic intermediaries and semiconductor producers in propagating foreign shocks, and suggest that the effects of globalization extend far beyond directly trading firms. Understanding how trade shocks differ by the sectors most affected, their upstream or downstream position, and the extent of intersectoral connectedness would yield a more complete picture of the

aggregate effects of globalization on Taiwan’s economy.

Related literature. This dataset opens several promising avenues for future research, particularly in the context of recent global disruptions and the renewed focus on shock propagation, supply chain resilience, and industrial policy. Taiwan’s production network is especially well-suited for studying these issues, given its central role in global value chains, the international prominence of its high-tech manufacturing sector, and its growing strategic importance in the geopolitical landscape.

First, Taiwan provides an ideal setting to study the effects of the U.S.-China trade war and the COVID-19 shock. By combining the B2B data with customs records, researchers can trace how Taiwanese firms were directly and indirectly affected by product-specific tariffs on Chinese exports imposed under the Trump administration. This setup also allows estimation of product-level price elasticities for both exporters and importers, and the analysis of heterogeneity based on firm-level characteristics such as age, size, and input intensity. When further combined with employee-employer matched data, the setting lends itself to studying labor turnover, reallocation, and mobility in response to global shocks (Komatsu and Dhyne, 2023; Dhyne et al., 2025), as well as the evolution of labor market power and earnings inequality (Huneus et al., 2025). Changes in supplier and buyer compositions following plausibly exogenous global events also offer a natural context to study network adjustment and supply chain resilience (Grossman et al., 2024a,b).

Second, given the skewness of Taiwan’s production network toward large manufacturers—many of whom are also key global players—it is natural to examine the role of superstar firms and the spillovers they generate through firm-to-firm linkages (OECD, 2025b). This speaks directly to recent work on superstar effects and network spillovers (Alfaro-Urena et al., 2022; Amiti et al., 2024), and also to research on endogenous network formation, where the architecture of firm connections can shape the magnitude and direction of spillovers (Huneus, 2019; Lim, 2018; Bernard et al., 2019; Huneus et al., 2025; Demir et al., 2024; Arkolakis et al., 2025). With the revitalization of industrial policy in many advanced economies—some aimed at reshoring production or attracting Taiwanese manufacturers—understanding the domestic role of these leading firms is a necessary first step. More broadly, as industrial policy increasingly intersects with national security considerations, Taiwan’s unique geopolitical position makes it a critical case for studying the supply chains of dual-use technologies and strategic inputs (Alekseev and Lin, 2026; Becko and O’Connor, 2025).

The research potential unlocked by proper use of this dataset extends well beyond the empirical exercises and literature review shown here. We hope this paper lowers the entry costs for future work, and contributes to a broader research agenda on production networks, trade, labor, firm dynamics, and related topics.

The rest of the paper is organized as follows. In Section 2, we describe the background of the value-added tax system in Taiwan and the firm-to-firm transaction data. In Section 3, we examine the credibility of the data sets, as well as the data cleaning and validation procedures. We then document several empirical regularities in Section 4, which researchers can compare with the results using network data from other countries. In Section 5, following [Dhyne et al. \(2021\)](#), we decompose the direct and indirect exposure of domestic firms to foreign shocks. We provide some final remarks for this technical paper in Section 6.

2 Data

2.1 Value-added Tax (VAT)

The two main data sources on domestic firm-to-firm transactions are the value-added tax (VAT) reports and the business-to-business (B2B) transaction data. The VAT data, including income statements, help researchers track firm-level sales revenues and operation costs. Starting from 1986, Taiwan’s tax authority has required that most sales revenues be taxed through VAT, with exceptions mainly applying to small businesses. Firms with monthly goods sales below 80,000 New Taiwan Dollars (NTD) (or service sales below 40,000 NTD) are exempt from VAT and are not included in our data.² Firms with monthly sales between 80,000 and 200,000 NTD and certain restaurants are subject to a non-VAT gross receipts tax, with sales estimated by the tax authority and taxed at a 1% rate.³

For businesses operating under the VAT system, a standard rate of 5% is levied on the value added at each stage of production and distribution. Some transactions, particularly those involving exports or firms in bonded zones—including bonded factories, warehouses, logistics centers, and science and technology parks—are zero-rated, as they bypass the domestic market. Zero-rated transactions include sales to bonded zone firms for operational use and direct exports by bonded zone firms.⁴

For firms complying with VAT, researchers observe separate values for business-to-business

²1 USD $\approx$ 30 NTD during the sample period. Starting January 1, 2025, Taiwan’s Ministry of Finance raised the business tax exemption threshold for small-scale businesses—from NTD 80,000 to NTD 100,000/month for goods sales, and from NTD 40,000 to NTD 50,000/month for services—to reflect cumulative CPI growth since 2007. An estimated 120,000 small businesses benefit. Please refer to this [website](#) for further details.

³Financial institutions are similarly excluded from the standard VAT system but subject to a distinct non-VAT business tax at higher rates.

⁴It is useful to distinguish VAT exemption from zero-rating. Under VAT, sellers collect output VAT from buyers and deduct input VAT paid on purchases. For zero-rated transactions, sellers apply a 0% VAT rate but may still claim deductions on input VAT paid on their own purchases. In contrast, VAT-exempt transactions are excluded from the VAT chain entirely: no output VAT is charged, and sellers cannot claim input VAT deductions on inputs related to exempt sales. Please refer to Article 7 (zero-rated) and Article 8 (VAT exemption) of the [Value-added and Non-value-added Business Tax Act](#).

(B2B), business-to-consumer (B2C), and export sales. Firms are required to record sales using distinct receipt formats corresponding to each transaction type. This institutional feature allows us to distinguish and compare B2B sales revenues reported in the tax data with the detailed firm-to-firm transaction dataset introduced below.

2.2 Business-to-business Transactions (B2B)

In addition to the VAT reports, the Ministry of Finance collects bimonthly B2B transaction records from 2013 onward. The unit of observation is the value of total transactions between each buyer-seller pair over a two-month period. Variables include total values for sales, returned sales, and taxes levied. It is worth noting that Taiwan imposes no minimum sales threshold for reporting over the two-month period. By linking seller IDs to firm tax IDs in the VAT system, researchers can, in principle, compare domestic B2B sales recorded in both sources to assess consistency.⁵ We describe this exercise in the following section.

2.3 Other Relevant Datasets

We further supplement the VAT and B2B datasets with several others for analyses in later sections. First, our analysis focuses on firms appearing in the business and company registry directory of Taiwanese formal entities under the Ministry of Finance.⁶ In addition to the unique entity identifier that allows us to merge across data sources and the industry code, it provides information on each firm’s date of registration, establishment, exit, and reopening at the month-year level. This helps researchers understand firm dynamics along the extensive margin.

Second, we use individual income tax (IIT) data to construct an employer-employee matched dataset. Each employee’s IIT record discloses the employer’s identifier, which can be linked to the firm’s tax ID, yielding firm-level payroll and headcounts. By applying a full-time minimum annual earnings threshold, we can further distinguish full-time from part-time employees, providing both a total employment count and a full-time-equivalent measure of firm size. We supplement these records with labor insurance enrollment data, which independently report the number of insured workers at each firm.⁷ The labor insurance

⁵Similarly, linking buyer IDs allows researchers to examine the consistency of firms’ input cost structures.

⁶Formal business entities are registered either as a company (公司登記, under the Company Act) or as a business (商業登記, under the Business Registration Act). The former applies to limited companies and corporations; the latter covers smaller sole proprietorships and partnerships. Regardless of type, all entities must complete a separate tax registration (稅籍登記), which generates the unique tax ID used to link records across our datasets. Businesses without a physical storefront whose monthly sales fall below 100,000 NTD for goods or 50,000 NTD for services are exempt from both tax registration and business tax. These entities are entirely outside the tax system and do not appear in our data.

⁷Only firms with more than five full-time employees are required to participate in the labor insurance

headcounts serve as a useful cross-check on the employment measures derived from the IIT data. Together, these sources allow researchers to characterize labor costs and to use employment—in addition to sales—as an alternative measure of firm size.

Lastly, while the VAT dataset discloses import and export information based on receipt types, we also combine the transaction-level customs records, which we aggregate to firm-level imports and exports by six-digit Harmonized System (HS) product code and trading partner country.⁸

3 Pre-analysis Data Preparation

3.1 Consistency of Sample

While Taiwan’s VAT system excludes firms with limited monthly operating revenue, there is no value threshold for reporting B2B transactions. In addition, although public entities are exempt from the VAT, the firms that sell to them still issue receipts for these transactions, so public entities appear as buyers in the network. As a result, some buyers or sellers—presumably smaller businesses or public entities—appear in the B2B transaction records but not in the VAT system. Using cross-sectional data from 2017, [Table 1](#) summarizes the coverage of firms appearing only in VAT, only in B2B, or in both datasets. Throughout this paper, we call a business entity with a unique tax identifier a firm.

There are 40,026 business entities with VAT registration that do not appear in any B2B transaction, either as a seller or a buyer; these businesses typically serve end markets directly through business-to-consumer (B2C) transactions or exports. The B2B dataset contains 1,469,417 unique firms, appearing as sellers, buyers, or both. Among these firms, 560,659 do not have formal VAT registration. These firms lack formal VAT registration either because their revenue falls below the mandatory registration threshold, or because they exclusively deal in VAT-exempt goods and services under Article 8 of the Business Tax Act (e.g., agricultural produce, medical services, educational activities, or government-related work) and are thus exempt from tax registration under Article 29. While they account for 38% of firms in the data, their combined B2B sales represent only 0.035% of Taiwan’s domestic firm-to-firm transactions. By contrast, firms appearing in both datasets account for over 99.9% of all firm-to-firm transactions in Taiwan. As these firms hold formal VAT registration, researchers can also observe their input and output information through the balance sheet attached to their tax filings. More importantly, because their VAT filings

program.

⁸[Kao and Liu \(2022\)](#) and [Kao et al. \(2024\)](#) also combine labor, corporate income tax, and trade data to document empirical patterns of Taiwanese firm dynamics. Their combined dataset covers the period from 2006 onward, whereas the B2B data used in this study begin in 2013.

break down annual sales into domestic B2B, domestic B2C and exports according to receipt types, we can compare the implied annualized B2B sales in the VAT data with the buyer-seller pair-level B2B data to assess consistency and data quality.

Table 1: Business Entities in B2B and VAT Datasets (2017)

	Number of Tax IDs	Network Sales	% of Total B2B Sales
In VAT Only	40,026	-	-
In B2B Only	560,659	7.68×10^9	0.035%
In Both	908,759	2.17×10^{13}	99.965%
Total	1,509,443	2.17×10^{13}	100%

Notes: Network Sales are denominated in New Taiwan Dollars (NTD). “In VAT Only” includes firms with formal VAT registration that do not participate in domestic B2B transactions, either as a seller or a buyer. “In B2B Only” includes firms appearing in the B2B records but without formal VAT registration. “In Both” includes firms present in both datasets.

3.2 Consistency of Sales Revenues

In principle, the firm-level VAT and buyer-seller pair-level B2B datasets should convey the same information about domestic firm-to-firm sales. Focusing on firms appearing in both datasets, we compare the annual net sales amount from each dataset and examine the consistency.⁹ Using data from 2017, we define firm-level discrepancies of domestic firm-to-firm sales between the two datasets as:

$$\delta_i \equiv \log_{10} \left(\frac{Sales_i^{VAT}}{Sales_i^{B2B}} \right) \quad (1)$$

For firms with perfectly matched sales values, the value of δ is 0. The distribution of δ is illustrated in [Figure 1](#), where panel (a) shows the full sample and panel (b) restricts to imperfect matches ($\delta \neq 0$). The bars display the histogram of δ values (corresponding to the scale on the left), whereas the black dashed line shows the cumulative B2B sales (scale on the right).

Overall, the firm-level B2B sales implied from the two datasets are highly consistent. The histogram of the full sample exhibits a spike around 0, where firm-level sales are identical in both datasets. Among imperfect matches, the distribution remains concentrated near zero, and the discrepancy is asymmetric: firms more commonly report higher sales in the VAT data than what is implied by the firm-to-firm transaction records. The first panel of [Table 2](#) presents the breakdown of the discrepancy distribution. With the full sample, we observe an 85% perfect match rate between the two datasets. Moreover, only 8% of firms have a discrepancy larger than 10% of their domestic B2B sales (i.e., $\delta_i \notin [\log_{10}(0.9), \log_{10}(1.1)]$).

⁹Net sales amount is defined as sales minus sales returns.

The match quality varies by industry. In the remaining panels of Table 2, we present the composition of δ for manufacturing, wholesale and retail, and hotel and restaurant industries. The match rate is highest among manufacturing firms, with only 14% of B2B sales in the sector contributed by firms with discrepancies exceeding 10%. In the wholesale and retail sector, while the share of firms in the “Outside 10%” category is larger than that for the full sample, their contribution to B2B sales (16%) is below that for the full sample (17%). The hotel and restaurant sector is the primary source of discrepancies: 46% of firms in this sector have discrepancies exceeding 10%, accounting for 48% of the sector’s B2B sales.

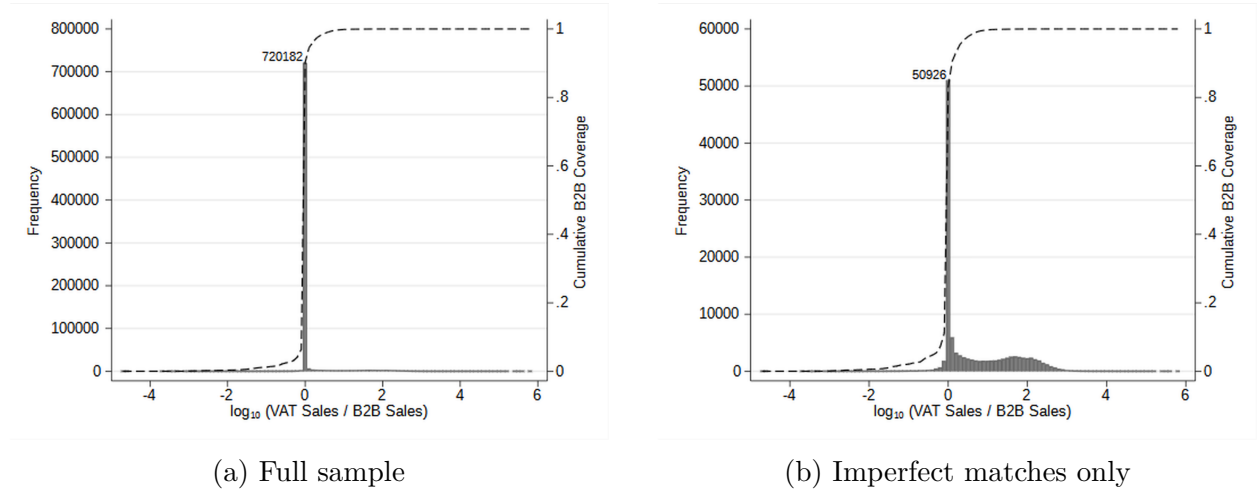


Figure 1: Discrepancies of sales between VAT and B2B datasets (2017)

Notes: The histogram (left y-axis) shows the distribution of discrepancies between the B2B and VAT datasets, as defined in Equation 1. The dashed line (right y-axis) indicates the cumulative B2B sales coverage. Panel (b) restricts to firms with imperfect matches ($\delta \neq 0$).

Causes of inconsistencies. There are two common sources of discrepancies. First, they may stem from decimal misreporting, as documented in several previous studies.¹⁰ Specifically, such errors would involve factors of 1,000 under the modern accounting system, or factors of 10,000 under the traditional Taiwanese accounting system. This would imply clusters of δ precisely at $\pm(3, 6, 9)$ or $\pm(4, 8, 12)$. However, we do not observe such patterns in our data.

Second, discrepancies may also arise from differences in the reporting of returned sales, since we compare net sales amounts (B2B sales net of returns). For example, consider a B2B transaction in December 2017, with the buyer returning part of the purchase in January

¹⁰For example, Dhyne et al. (2015) correct Belgian B2B transaction values by meticulously inspecting misreporting issues, such as decimal errors and wrongly coded negative transactions, and by imputing missing transactions when gaps are observed. In later vintages of the Belgian data, Dhyne et al. (2023) adopt a more data-driven approach, recursively correcting transaction values and total sales inferred from other datasets to achieve internal consistency.

Table 2: Distribution of Match Types

Sample	Match type	% of firms	% of B2B sales
Full Sample	Perfect	85.41	42.31
	Within 1%	4.20	26.38
	Within 10%	2.16	14.26
	Outside 10%	8.23	17.05
Manufacturing (8–34)	Perfect	89.88	36.41
	Within 1%	5.45	32.98
	Within 10%	2.40	16.75
	Outside 10%	2.28	13.85
Wholesale and Retail (45–48)	Perfect	82.67	42.49
	Within 1%	4.59	27.34
	Within 10%	2.25	13.90
	Outside 10%	10.50	16.27
Hotel and Restaurant (55, 56)	Perfect	48.31	34.99
	Within 1%	3.56	10.05
	Within 10%	2.00	6.82
	Outside 10%	46.12	48.14

Notes: Categories are mutually exclusive. “Within 1%”: $\delta_i \in [\log_{10}(0.99), \log_{10}(1.01)]$, excluding perfect matches. “Within 10%”: $\delta_i \in [\log_{10}(0.9), \log_{10}(1.1)]$, excluding within 1%. 2-digit industry codes in parentheses.

2018. In this case, the seller’s 2017 B2B revenue would align across the two datasets, but the treatment of the return could differ. The annual VAT report would attribute the return to 2017 sales activity, while the B2B data might record it in 2018, reflecting the exact timing of the transaction. This mismatch generates differences in firms’ annual net domestic sales across datasets. We observe several such cases in our investigation. However, because both datasets are aggregated at the bimonthly rather than transaction level, we cannot systematically match returns to their corresponding sales to correct these discrepancies.

Systematic correction of these discrepancies is not feasible without invoice-level data. In the next subsection, we describe how we construct firm-level sales and costs that are internally consistent for the subsequent analysis.

3.3 Data Cleaning and Baseline Sample Construction

Since the discrepancies are not driven by decimal misreporting, such corrections are unnecessary. It is also difficult to systematically correct discrepancies arising from differences in the timing of reported returns. Consequently, we do not impose any ad hoc restrictions or corrections.

We construct an internally consistent firm-level dataset of total sales and costs by lever-

aging several additional sources. Following [Dhyne et al. \(2021\)](#), we define a firm’s total sales as the sum of domestic firm-to-firm sales from the B2B dataset, sales to non-firm consumers from the VAT dataset, and export values from customs data.¹¹ Total costs are constructed by summing network inputs (domestic B2B purchases), import values from customs records, and labor costs from aggregating labor income in the personal income tax.¹² We can define the total sales and costs as follows:

$$\begin{aligned}\text{Total Sales}_j &= \text{B2BSales}_j + \text{B2CSales}_j + \text{Exports}_j \\ \text{Total Costs}_j &= \text{B2BPurch}_j + \text{Imports}_j + \text{LaborCost}_j\end{aligned}\tag{2}$$

By construction, each firm’s total sales and costs are no smaller than any individual component.

We adopt the selection process of [Dhyne et al. \(2021\)](#) to obtain the cleaned sample for the baseline analysis in the remainder of this paper. First, we exclude firms belonging to public services and financial sectors.¹³ Second, we exclude firms with zero total sales or missing labor costs. Once we remove these firms, we also remove them as buyers or sellers from the B2B dataset. With the updated network, we recalculate the B2B sales and purchases and update total sales and total costs. Throughout this process, some firms become zero-total-sale firms, as B2B sales are their only source of income and they lose all customers in the previous round of cleaning. We drop these firms and repeat these steps until all remaining firms in the sample have (1) positive total sales and (2) positive labor costs.

[Table 3](#) shows the coverage of key variables in the full sample, the cleaned sample, and the aggregate statistics published by Taiwan’s official statistical bureau. We collect aggregate statistics from the Directorate General of Budget, Accounting and Statistics (DGBAS). First, the full sample closely tracks the aggregate statistics, with the exception of labor costs.¹⁴ Second, focusing on the comparison between the full and cleaned samples, we find that while the number of firms in our cleaned sample accounts for only 37% of that in the full sample, we capture over 83% of sales and costs, with labor costs again being the exception. Further

¹¹An alternative definition of export sales is the sum of zero-rated sales net of returns reported in the VAT.

¹²We first aggregate individual income tax (IIT) records by ID of the employer’s business entity to obtain total payroll at the firm level. Payroll figures from the IIT data are generally more accurate, as employees have incentives to verify reported values to avoid excessive taxation; thus, we use IIT as our primary source.

¹³In Taiwan’s two-digit Standard Industrial Classification (SIC), these include sector codes 83–93 (public services, national defense, education, etc.) and 64–66 (financial). See the [Ministry of Finance website](#) for the full classification.

¹⁴The discrepancy arises because the DGBAS aggregate measure of compensation of employees (受僱人員報酬) includes employer-side social insurance contributions (雇主社會保險捐), whereas our labor cost measure is constructed by aggregating individual income tax records, which capture regular and nonregular wages and salaries paid to employees but not the employer-side contributions. In Taiwan, these contributions include national health insurance (全民健康保險), labor insurance (勞工保險), employment insurance (就業保險), and the new labor pension (勞退新制).

inspection reveals that the difference in labor costs is due to the exclusion of zero-sales firms after the sector selection; these dropped firms appear only as buyers in the B2B dataset and as employers in the personal income tax data, with no record in tax registration or the VAT system.¹⁵ The most reasonable explanation is that these firms provide goods and services that are fully exempt from VAT, most likely in the public sectors (e.g., schools and hospitals).

Table 3: Data Coverage of Aggregate Statistics, Raw Sample, and Cleaned Sample

	# Tax IDs	Total Sales	B2B Sales	Export Sales	Import Costs	Labor Costs	Employ.
Aggregate [†]	1,545,539	39.10	—	9.60	7.83	8.04	1.14×10^7
(1) Full	1,509,443	39.73	21.67	8.88	7.15	5.22	1.07×10^7
(2) Cleaned	561,566	33.68	18.02	8.50	6.75	3.59	0.81×10^7
(2) / (1)	37.20%	84.89%	82.95%	95.75%	84.41%	68.77%	75.57%

Notes: All monetary values are denominated in trillions of New Taiwan Dollars (NTD $\times 10^{12}$).

[†]Aggregate statistics are sourced from the National Statistics of R.O.C. (Taiwan) administered by the Directorate General of Budget, Accounting and Statistics (DGBAS). Available at the [DGBAS Statistical Database](#). The “Employ.” column reports the number of unique employee IDs, which includes both full-time and part-time workers.

4 Stylized Facts of Taiwan’s Production Network

Using the cleaned data as the baseline sample, we present several stylized facts about Taiwanese domestic production networks. For cross-sectional patterns, we use data from the year 2017 to avoid potential structural changes in production networks due to the U.S.-China trade war since 2018 and the COVID pandemic shock since 2020. While we use the cleaned sample to present the results in the rest of this paper, we also report summary statistics of the full sample in Subsection [A.1](#) for comparison.

¹⁵This distinguishes them from the public-service and financial firms excluded in the first round, which do appear in tax registration (and thus have an identifiable Standard Industrial Classification (SIC) code) and may also file VAT returns for any taxable goods or services they provide alongside their primary exempt activities. The firms that contributed to the difference in labor costs, by contrast, are entirely outside the tax registration system—consistent with Article 29 of the Business Tax Act, which exempts businesses exclusively dealing in VAT-exempt goods and services from the registration requirement. Without a tax registration record, their industry codes are unknown, and they cannot be caught by the sector-based filter in the first round.

Fact 1. Domestic B2B Transactions Have Been Steadily Growing

Figure 2 shows that domestic transactions increased steadily from 2013 to 2022 along both the extensive margin (the number of firm-to-firm linkages, Figure 2a) and the intensive margin (the total value of firm-to-firm sales, Figure 2b). Over this period, the number of buyer-seller pairs rose from about 33 million to more than 45 million. At the same time, total domestic firm-to-firm sales grew from roughly 17 trillion NTD to nearly 24 trillion NTD. For comparison, nominal GDP increased from 15 to 23 trillion NTD, implying a stable network sales to GDP ratio around 1.1.¹⁶

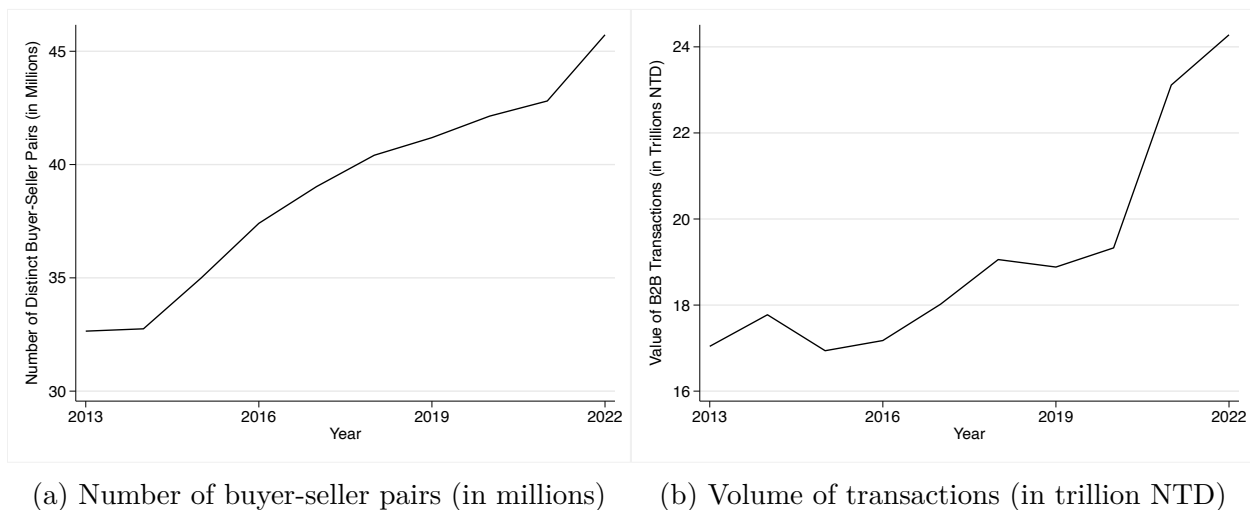


Figure 2: Development of Taiwanese domestic supply chains, 2013–2022

Figure 3 further displays the trend of average firm-level network cost and sales shares. Figure 3a and Figure 3b both show a relatively stable role for B2B transactions over time, with network purchases accounting for slightly above 60% of total costs and network sales accounting for roughly 50% of total revenues. All patterns suggest a balanced growth of domestic network activities alongside Taiwan’s aggregate economic progress.

Fact 2. Domestic Networks are Skewed toward High-tech Manufacturing Sectors

The aggregate time trend easily masks substantial heterogeneity across sectors and firms. In practice, the distribution of network activities is highly skewed. According to the summary statistics in Table 4, the average firm had 69.50 domestic buyers and 69.50 domestic suppliers,

¹⁶The post-pandemic increase in domestic transaction volume since 2020 may reflect a localization of production in response to disruptions in global trade. We leave a detailed investigation of firms’ network behavior in response to the pandemic for future research.

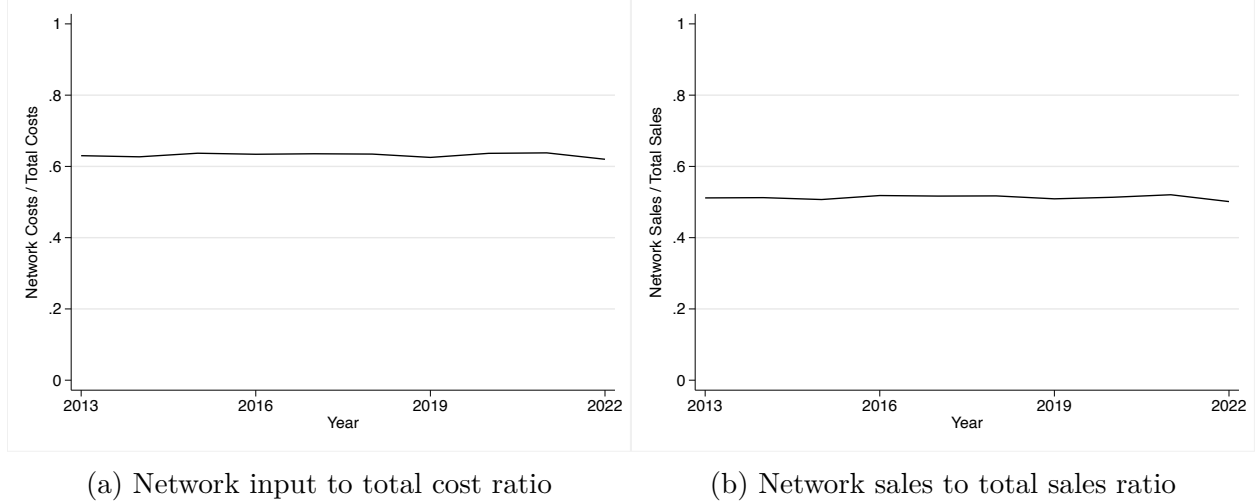


Figure 3: Output and Input Share of Domestic Network, 2013–2022

Notes: The total costs and total sales figures are constructed to maintain internal consistency. Specifically, we adhere to the methodology outlined in [Dhyne et al. \(2021\)](#) and define the total cost of a firm as the sum of domestic inputs, import costs, and labor costs. Similarly, we define total sales as the sum of sales to domestic firms, sales to customers, and exports. This ensures that the costs and sales shares of any component are bounded between 0 and 1.

Table 4: Summary statistics for firm-to-firm transactions in 2017

	Mean	Std.Dev.	Percentiles			
			25th	50th	75th	95th
Link level (N = 39,026,875)						
Transaction amount (in thousands NTD)	461.72	40885.11	1.21	5.85	40.00	724.85
Firm level (N = 561,566)						
Number of domestic buyers	69.50	649.12	3	10	35	199
Number of domestic suppliers	69.50	119.14	18	40	80	219
Domestic network sales (in millions NTD)	32.09	731.01	0.57	3.27	12.07	74.24
Domestic input purchases (in millions NTD)	32.09	766.54	0.55	2.43	9.14	70.70

Notes: This table uses the baseline cleaned sample in 2017. The link-level panel covers all unique domestic buyer–seller pairs; the transaction amount is the total annual value of transactions between each pair, denominated in thousands of New Taiwan Dollars (NTD). The firm-level panel covers all firms appearing as buyers or sellers. Number of domestic buyers (suppliers) is the count of unique downstream customers (upstream suppliers) for each firm in the domestic B2B network. Domestic network sales (input purchases) is the total annual value of a firm’s domestic B2B sales (purchases), denominated in millions of NTD. The mean number of buyers equals the mean number of suppliers, and mean network sales equals mean input purchases, as each domestic transaction is simultaneously recorded as a sale by the seller and a purchase by the buyer.

with 32.09 million NTD in both domestic sales and purchases. However, the median firm had only 10 domestic buyers and 40 domestic suppliers, with median domestic sales of 3.27 million NTD and median domestic purchases of 2.43 million NTD. This stark divergence

between mean and median values points to a small group of key firms—likely in sectors with greater network centrality—that disproportionately drive domestic linkages and sales. In the appendix, we report summary statistics for firm-to-firm transactions in 2013 (Table A.3) and 2022 (Table A.4), the start and the end of our data span, for comparison.

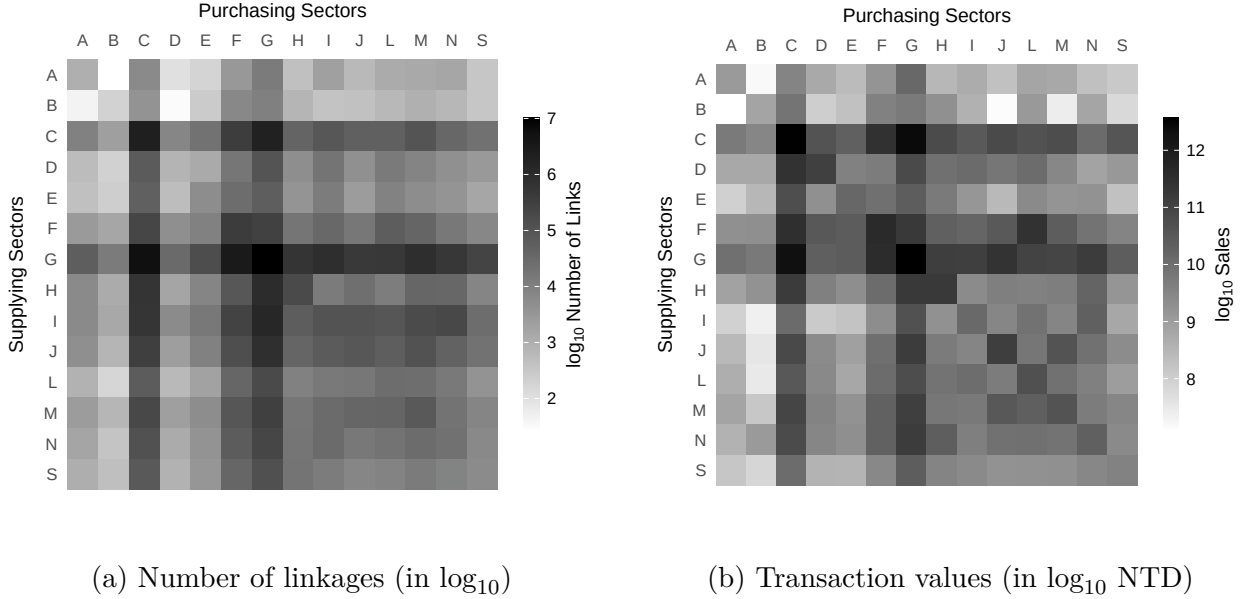


Figure 4: Inter-sectoral transactions (2017)

Notes: The sectoral codes follow the 9th edition of the Standard Industrial Classification. Table A.5 provides the definition for each sector code.

Sectoral network contributions. To understand the importance of each sector in domestic B2B transactions, we visualize inter-sectoral transactions as heatmaps in Figure 4. The left panel (Figure 4a) presents a heatmap of the number of linkages from the supplying sector (vertical axis) to the purchasing sector (horizontal axis), while the right panel (Figure 4b) shows the corresponding heatmap of transaction values. Table A.5 in the appendix provides descriptions of each sector code and reports the number of firms in each sector. In Taiwan, the manufacturing sector (code C) and the wholesale and retail trade sector (code G) are the major suppliers and purchasers, both in terms of the number of buyer-seller pairs and the total transaction value. In addition, the construction sector (code F) also plays a significant role within domestic networks.

In Table 5, we further break down the firm size and network linkage distributions by sector, where firm size is measured by total sales as constructed in Section 3.3. The first column lists the industries of interest, the second shows the share of firms in each sector; the total sales columns display the distribution of firm-level total sales (summing B2B, B2C,

Table 5: Firm size distribution of networks by industry, 2017

	All firms in 2017 (N = 561,566)					
	Share of firms	Total Sales (Mil. NTD)			Mean # of links	
		(%)	Mean	p50	p95	Buyers
<i>Economy-wide</i>						
All Industries	100	62.10	6.73	122.61	69.50	69.50
<i>Broad sectors</i>						
Agriculture & mining	0.43	35.67	7.10	119.12	25.71	56.10
Manufacturing	18.85	140.20	10.07	243.85	47.76	95.73
Utility & construction	13.56	37.59	6.12	79.97	20.55	64.74
Wholesale & retail	47.15	45.19	6.69	112.50	88.17	63.52
Service	20.01	45.54	4.99	87.93	80.10	62.14
<i>Electronics manufacturing</i>						
Electronic parts & components	0.86	902.20	17.87	1564.19	46.17	158.27
Computers, electronic & optical	0.43	316.67	15.00	970.78	61.24	167.80
<i>KTI industries</i>						
High-tech manufacturing	1.35	693.25	17.68	1278.59	52.29	169.81
Medium-high-tech manuf.	4.15	157.30	12.22	342.26	53.33	123.42
High-tech KI service	1.72	106.78	4.00	172.21	174.37	79.04
KI market service	5.21	35.24	3.60	70.51	36.32	59.49

Notes: The sample consists of all firms in the baseline cleaned sample ($N = 561,566$) in 2017. Share of firms reports each sector’s fraction of all firms in the sample. Total Sales is the sum of domestic B2B sales, business-to-consumer (B2C) sales, and export sales, denominated in millions of New Taiwan Dollars (NTD), as defined in Subsection 3.3. Buyers (Suppliers) reports the mean number of unique domestic downstream customers (upstream suppliers) per firm in the B2B network. Broad sector groupings follow Taiwan’s SIC: Agriculture & Mining includes sectors A–B; Manufacturing is sector C; Utility & Construction covers sectors D–F; Wholesale & Retail is sector G; Service covers sectors H–J and L–N, S. Electronics manufacturing comprises electronic parts & components (two-digit SIC code 26) and computers, electronic & optical products (code 27). KTI denotes knowledge- and technology-intensive industries, classified following Eurostat’s definitions; see Table A.6 for details.

and export sales), and the final set of columns reports the average number of upstream and downstream linkages.

The second panel summarizes statistics across broadly defined sectors, using “sections” from Taiwan’s SIC. Manufacturing firms, which have the largest average firm size, represent about 19% of all firms but are major contributors to domestic network activities and typically have more domestic suppliers than the average Taiwanese firm (mean suppliers: 95.73 vs. 69.50). By contrast, wholesale and retail firms account for 47% of all firms in domestic supply chains; though smaller on average, they serve roughly 27% more domestic downstream partners than the typical firm (mean customers: 88.17 vs. 69.50).

In the third panel, we focus on the electronics manufacturing industries, including elec-

tronic components (two-digit SIC code 26) and computer and electronic products (two-digit SIC code 27). Firms in these critical sectors have a larger median size, with even larger mean sales far exceeding the national average, likely driven by firms in the right tail of the distribution. Within networks, they serve as major customers for many domestic suppliers, with the mean number of suppliers roughly 2.4 times the nationwide average.

The last panel presents descriptive statistics for knowledge- and technology-intensive (KTI) industries classified by Eurostat.¹⁷ As detailed in Table A.6, the high-tech and medium-high-tech manufacturing industries correspond to pharmaceuticals, electronics, and machinery. These KTI manufacturing sectors account for only 5.5% of all firms in the network but are critical downstream customers within domestic supply chains. In contrast, the high-tech and knowledge-intensive market services—such as film and music production, broadcasting, telecommunications, and broader information and communications technology (ICT) services—play a key role as domestic suppliers, averaging 174 customers each. Finally, firms in knowledge-intensive market services are smaller on average and play a less important role in our baseline network.

Distribution of network in-degree and out-degree. Figure 5 plots the cumulative distributions of firm-level in-degree (number of suppliers) and out-degree (number of customers), where the cumulative density equals one when the degree (x-axis) is 1. Fewer than 10% of firms have more than 100 suppliers or customers, highlighting the skewness of the network. We approximate these distributions with Pareto distributions. The estimated shape parameters are 1.90 for in-degree and 1.11 for out-degree. As both estimates fall strictly between 1 and 2, the results imply that aggregate volatility decays at a rate slower than $\sqrt{n}$, where n is the number of firms in the network (Gabaix, 2011; Acemoglu et al., 2012).

Fact 3. Firms concentrate transactions on few customers and suppliers

Within each firm, network activities are also unevenly distributed. For each firm, we compute the share of domestic B2B sales accounted for by the top 1 and top 5 buyers, and the share of domestic B2B purchases accounted for by the top 1 and top 5 suppliers. We plot the distribution of top partner network contribution in Figure 6. For an average firm, the top 5 customers account for 81% of total B2B sales, while the top 5 suppliers account for 77% of total B2B purchases. This suggests that firms concentrate transactions on a small number of key partners in the supply chain.

¹⁷Lastly accessed on May 22, 2026 via https://ec.europa.eu/eurostat/cache/metadata/Annexes/htec_esms_an_3.pdf for Eurostat’s definitions for KTI, and <https://service.mof.gov.tw/public/Data/statistic/std/en/index.html#rev=9> for the corresponding Taiwan’s SIC divisions (9th revision).

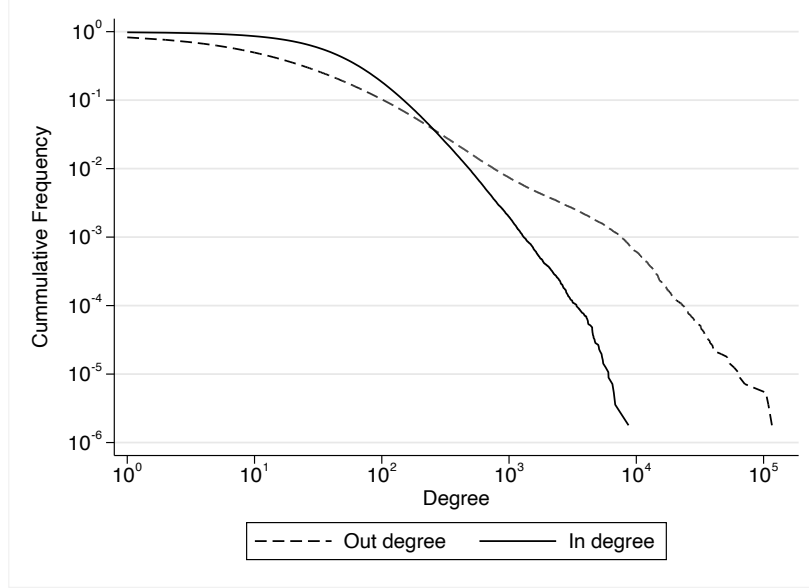


Figure 5: Degree Distribution of Taiwanese Production Network, 2017

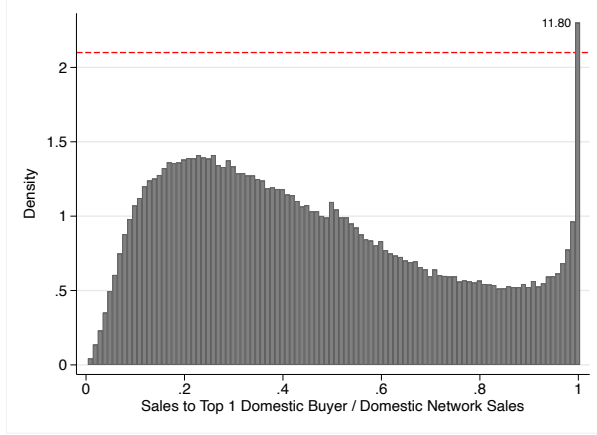
Notes: The figure plots the cumulative degree distribution of the Taiwanese production network. The horizontal axis represents the degree (in log scale), while the vertical axis represents the fraction of firms with degree exceeding the corresponding value.

Taken together, the descriptive evidence in this section reveals three salient features of Taiwan’s domestic production network. First, the network has grown steadily over the sample period, with the number of active firm-to-firm pairs rising by more than a third and the total value of B2B transactions growing by over 40% between 2013 and 2022. Second, the network is highly skewed in the cross-section: a small number of large firms—concentrated in high-tech manufacturing industries—account for a disproportionate share of domestic linkages and transaction values. Third, this concentration extends within firms as well: for a typical firm, the top five buyers and suppliers together account for the vast majority of its domestic B2B sales and purchases, implying that firm-level outcomes are tightly linked to a handful of key partners. These results add Taiwan to the growing literature documenting empirical regularities of production networks across many countries (Bacilieri et al., 2026).

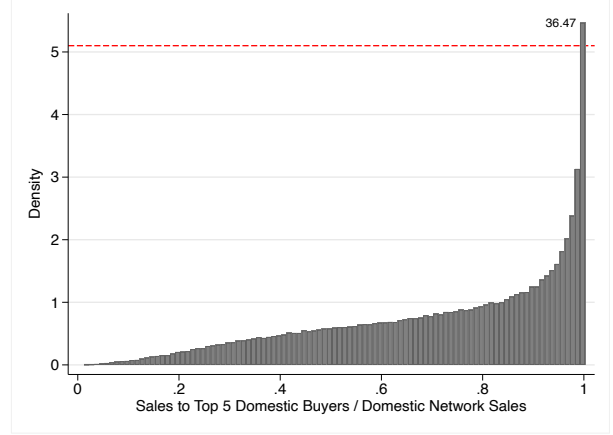
The skewness of Taiwan’s production network—toward wholesale and retail middlemen on the buyer side, and toward strategic semiconductor manufacturing on the supplier side—has direct implications for how international shocks propagate through the domestic economy, as we show in the following section.

5 Indirect Trade through Domestic Supply Chains

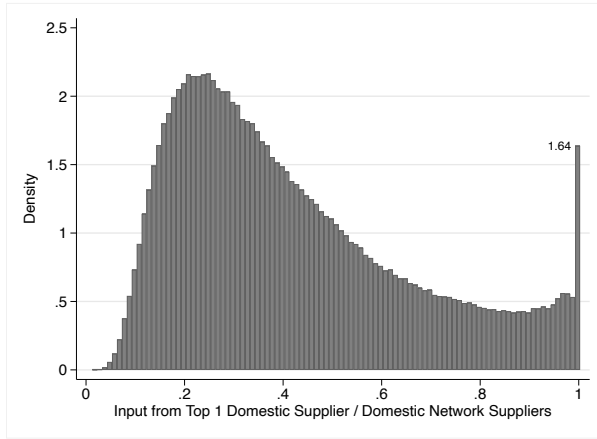
The interconnectedness of firms in domestic supply chains, along with the skewness of the network and the importance of key partners and key sectors, suggests that a disruption in



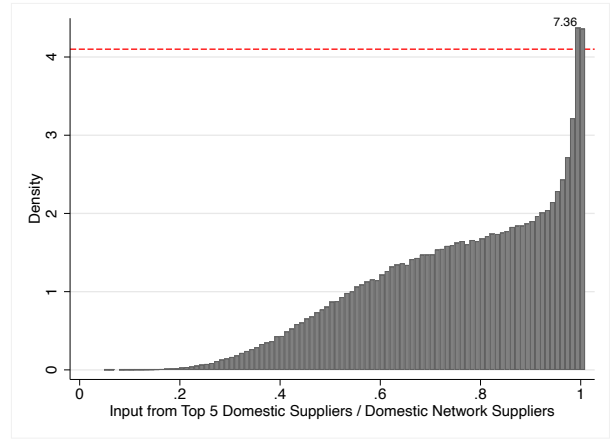
(a) Sales share to top 1 buyer



(b) Sales share to top 5 buyers



(c) Inputs share from top 1 supplier



(d) Inputs share from top 5 suppliers

Figure 6: Within-firm concentration of sales and inputs across buyers and suppliers (2017)

Notes: Using the data in 2017, these figures display the distribution of firm-level B2B sales shares accounted for by the top 1 and top 5 customers (first row), as well as the distribution of firm-level B2B purchases coming from the top 1 and top 5 suppliers (second row). See Figure 3 in [Dhyne et al. \(2023\)](#) for a direct comparison with the case of Belgium.

one part of the system can lead to significant aggregate outcomes through the propagation of shocks. Taiwan is a small open economy highly integrated in high-tech global value chains. The increasing geopolitical risks in the region raise a natural policy concern: what if Taiwan faces major disruptions in the distribution (retail and wholesale) and semiconductor sectors? How will domestic firms—connected to companies that directly import or export—be affected?

In this section, we demonstrate how domestic firms—though not direct importers or exporters—can nonetheless be affected by international market fluctuations through domestic supply chains. To do so, we combine B2B transaction data with customs records to calculate each firm’s direct and indirect exposure to foreign markets, following the approach

of [Dhyne et al. \(2021\)](#). The results presented here use the sample from 2017.

5.1 Implementation

Our cleaned data represent a self-contained network system in which the costs and sales of each firm can be traced to either foreign or domestic sources, and the share of each cost and sales term falls strictly between 0 and 1. For each firm j , it is straightforward to construct the direct import cost share (s_{Fj}) and direct export sales share (r_{jF}) as follows:

$$s_{Fj} = \frac{\text{Imports}_j}{\text{B2BPurch}_j + \text{LaborCost}_j + \text{Imports}_j}, \quad r_{jF} = \frac{\text{Exports}_j}{\text{B2BSales}_j + \text{B2CSales}_j + \text{Exports}_j}. \quad (3)$$

The denominators represent, respectively, the total input costs and total sales of each firm.¹⁸

We now explain how to construct the firm-level *total import share* (s_{Fj}^{Total}) and *total export share* (r_{jF}^{Total}). To compute the total import share for each firm j , we first calculate the share of firm j 's total input costs sourced from each supplier i , denoted by s_{ij} . The first-degree indirect import exposure of firm j is then obtained by weighting each supplier i 's direct import share by s_{ij} . Following the same logic, we calculate the second- and higher-degree indirect import shares as follows:

$$\begin{aligned} s_{Fj}^{\text{Total}} &= \underbrace{s_{Fj}}_{\text{Direct}} + \underbrace{\sum_{i \in Z_j^D} s_{ij} s_{Fi}}_{\text{1st}} + \underbrace{\sum_{i \in Z_j^D} s_{ij} \sum_{k \in Z_i^D} s_{ki} s_{Fk}}_{\text{2nd}} + \underbrace{\sum_{i \in Z_j^D} s_{ij} \sum_{k \in Z_i^D} s_{ki} \sum_{\ell \in Z_k^D} s_{\ell k} s_{F\ell}}_{\text{3rd}} + \underbrace{\dots}_{\text{4th and higher}} \\ &= \underbrace{s_{Fj}}_{\text{Direct}} + \sum_{i \in Z_j^D} s_{ij} \left[s_{Fi} + \underbrace{\sum_{k \in Z_i^D} s_{ki} (s_{Fk} + \dots)}_{s_{Fi}^{\text{Total}}} \right] \end{aligned} \quad (4)$$

Similarly, the total export share for each firm j can be derived as

$$r_{jF}^{\text{Total}} = r_{jF} + \sum_{i \in W_j^D} r_{ji} \left[r_{iF} + \underbrace{\sum_{k \in W_i^D} r_{ik} (r_{kF} + \dots)}_{r_{iF}^{\text{Total}}} \right] \quad (5)$$

¹⁸As discussed in Section 3.3, our labor cost measure captures wages and salaries from individual income tax records but excludes employer-side social insurance contributions (e.g., health insurance, labor insurance, employment insurance, and labor pension). Since labor cost enters the denominator of the direct import share s_{Fj} , understating it leads to an upward bias in import shares throughout our analysis. The direct and total export shares, whose denominators contain only sales components, are not affected by this measurement issue.

Here Z_j^D denotes the set of domestic suppliers of firm j , and W_j^D denotes the set of domestic customers of firm j . As shown in the second line of Equation 4, firm j 's total import share equals the sum of its direct import share and the cost-share-weighted total import shares of its suppliers. In matrix form, this relationship can be expressed as

$$\mathbf{s}_F^{\text{Total}} = \mathbf{s}_F + \mathbf{S}' \mathbf{s}_F^{\text{Total}},$$

where $\mathbf{s}_F = (s_{F1}, \dots, s_{Fj}, \dots)'$ is the vector of firm-level direct import shares and $\mathbf{S} = (s_{ij})_{i,j \in \mathcal{N}}$ is the cost-share matrix, with each entry s_{ij} representing the share of firm j 's total input costs sourced from firm i . Here $\mathcal{N}$ denotes the set of firms in the network.

Similarly, firms' total export shares are defined as

$$\mathbf{r}_F^{\text{Total}} = \mathbf{r}_F + \mathbf{R} \mathbf{r}_F^{\text{Total}},$$

where $\mathbf{r}_F$ is the vector of direct export shares, $\mathbf{R} = (r_{ij})_{i,j \in \mathcal{N}}$ is the revenue-share matrix, and r_{ij} denotes the share of firm i 's total revenue derived from sales to firm j .¹⁹ By rearranging the above expressions, $\mathbf{s}_F^{\text{Total}}$ and $\mathbf{r}_F^{\text{Total}}$ can be obtained by premultiplying the vectors of direct import and export shares by the corresponding Leontief inverses with respect to input costs and sales:

$$\begin{aligned} \mathbf{s}_F^{\text{Total}} &= (I - \mathbf{S}')^{-1} \mathbf{s}_F \\ \mathbf{r}_F^{\text{Total}} &= (I - \mathbf{R})^{-1} \mathbf{r}_F. \end{aligned} \tag{6}$$

5.2 Distribution of Total Import and Export Shares

Figure 7 displays the distributions of total import shares (left) and total export shares (right) using data from 2017. For comparison, the histograms of direct import and export shares are shown in blue, and those of total trade shares are shown in light grey. Fewer than 15% of firms in the sample participate in direct import or export. Weighting by firm-level total sales, the average direct import share is 21.5% and the average direct export share is 24.3%, while the average total import share rises to 52.5% and the total export share to 46.0%. In other words, even firms without direct links to foreign partners are substantially exposed to international market fluctuations through their domestic supply chain connections, on both the input and output sides.

In Figure 8, we further decompose the contributions of first- through fourth-degree indirect trade to the total trade share. The left column presents the results for total import

¹⁹The convention for notation follows the direction of transaction flow. Accordingly, s_{ij} denotes sales from firm i to firm j as a share of j 's total input costs, while r_{ij} denotes sales from firm i to firm j as a share of i 's total revenue. The subscript F represents "foreign," so s_{Fi} refers to imports (purchases from abroad by firm i), and r_{iF} denotes exports (sales abroad by firm i).

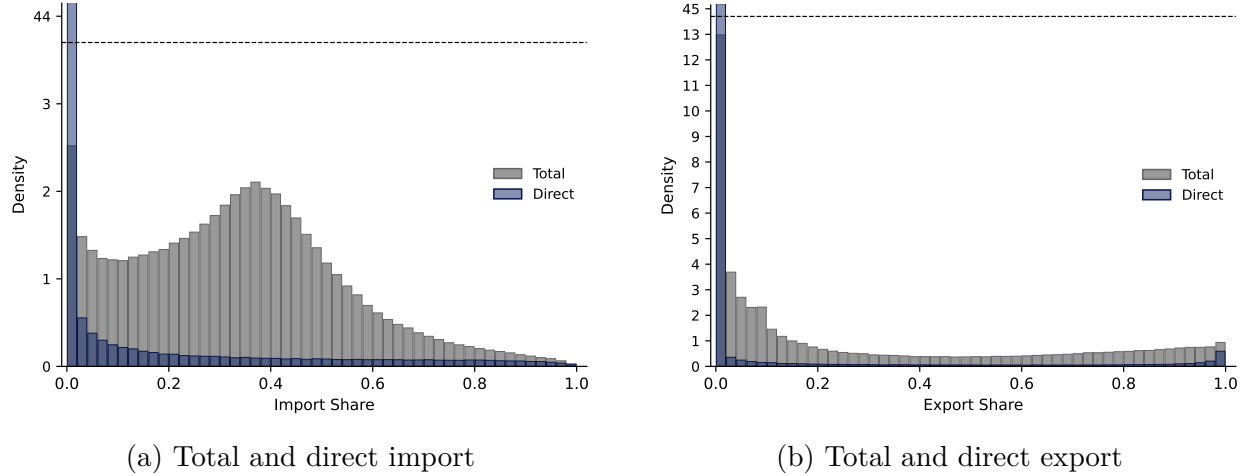


Figure 7: Total and Direct Trade, 2017

Notes: These figures plot the distribution of direct and total trade shares computed from Equation 6. The shaded grey histograms show the firm-level total trade shares, while the slate-colored histograms show the direct trade shares. See Figure 1 in Dhyne et al. (2021) for a comparison with the case of Belgium.

shares, while the right column shows those for total export shares. The findings suggest that most firms are no more than four degrees of separation away from international upstream or downstream markets. Notably, the transactional distance between firms and foreign markets is shorter on the sales side than on the cost side, as the first- through fourth-degree indirect exports account for the majority of a firm’s total exports, as opposed to the case for imports. This pattern also contrasts with the findings for Belgium in Dhyne et al. (2021), where first-degree indirect trade already accounts for the bulk of firms’ indirect trade exposure. In Taiwan, the prominent role of wholesalers and retailers as trade intermediaries—documented in Section 4—means that a typical firm’s domestic trading partners are often themselves intermediaries rather than direct importers or exporters. As a result, indirect trade exposure is spread more evenly across higher degrees of separation.

5.3 Indirect Trade through Wholesale and Retail Sectors

As discussed in Section 4, Taiwan’s production networks are highly skewed toward the retail, wholesale, and high-tech manufacturing sectors. It is particularly relevant, especially in the current geopolitical climate, to consider the role of these industries in transmitting foreign market fluctuations through the domestic supply chain.

We first consider a scenario in which we shut down the direct imports and exports of wholesalers and retailers (two-digit industry codes 45–48) and recalculate the total trade shares. The results are presented in Figure 9. The cluster around 0.4 in the distribution of import shares, observed earlier in Figure 7, disappears once the direct imports of wholesalers

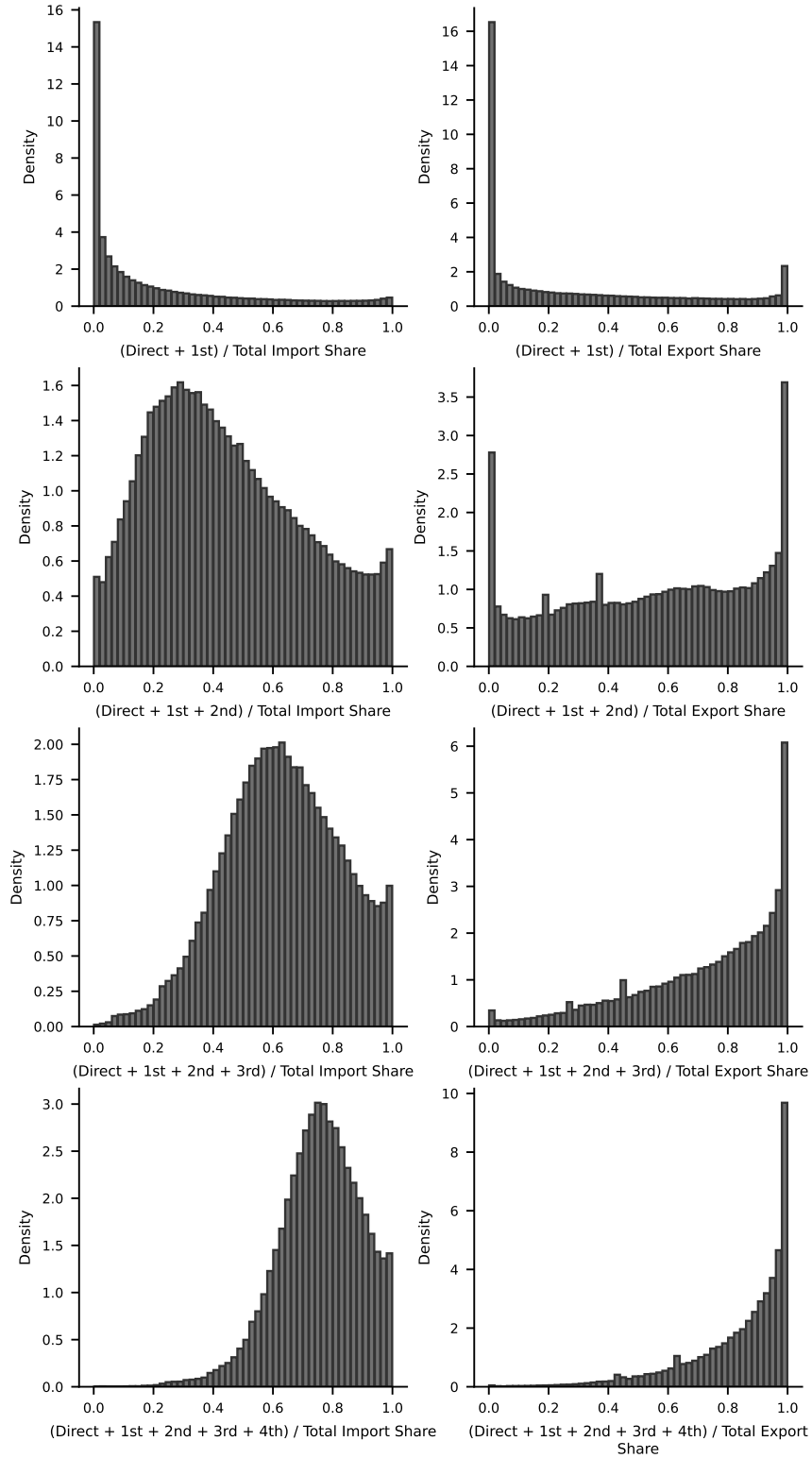


Figure 8: Decomposition of total import and export

Notes: The figure plots the first- through fourth-degree indirect trade as a share of total trade, as defined in [Equation 4](#) and [Equation 5](#). The left panel presents the results for imports, while the right panel presents those for exports. From the first to the fourth row, we display histograms of the cumulative contributions of the first through fourth degrees of indirect trade.

and retailers are excluded. The average total import share, weighted by firm-level sales, drops from 52.5% to 30.5%, suggesting that a majority of firms rely on these sectors to access international commodities. Likewise, the heavy tail in the distribution of export shares vanishes when the direct exports of wholesalers and retailers are set to zero, with the sales-weighted average total export share falling from 46.0% to 34.0%, indicating that most firms export indirectly through these sectors. This finding contrasts with [Dhyne et al. \(2021\)](#), where excluding wholesalers and retailers from direct trade does not materially affect the total trade share for the Belgian production network. Although both Taiwan and Belgium are small open economies that rely heavily on trade, wholesalers and retailers play a more prominent role in mediating international transactions in Taiwan.

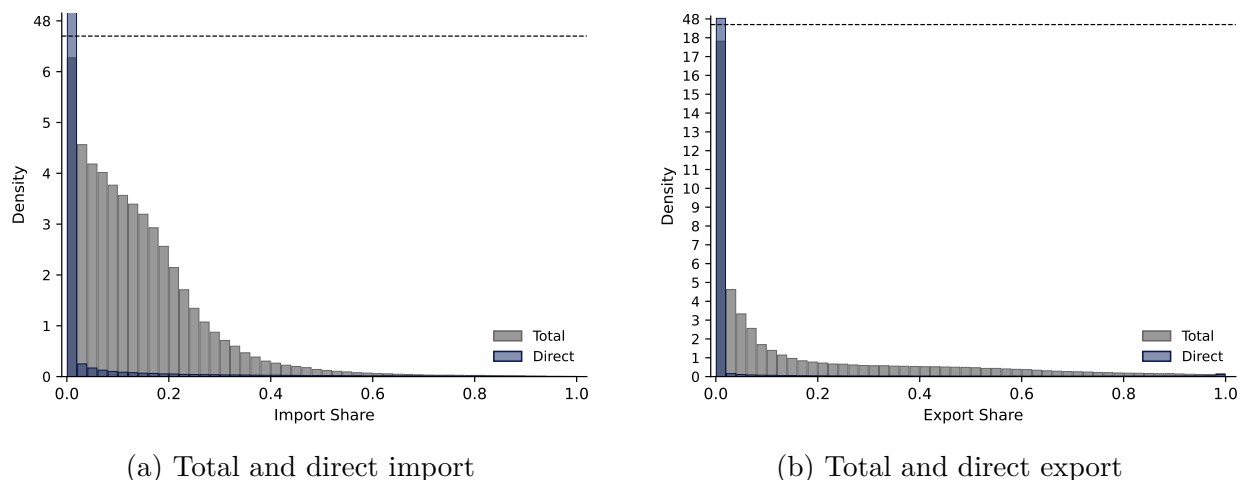


Figure 9: Total Import and Export Excluding Wholesalers and Retailers, 2017

Notes: These figures show the total and direct import and export shares after excluding the direct international trade of wholesalers and retailers. The shaded grey histograms depict firm-level total trade shares, while the slate-colored histograms depict direct trade shares.

5.4 Indirect Trade through Semiconductor Products

Next, we consider a trade disruption affecting semiconductor inputs imported from abroad and final semiconductor goods exported to foreign customers, following the HS code classification of [OECD \(2025a\)](#) summarized in [Table A.7](#).²⁰ The results are presented in [Figure 10](#). A striking asymmetry emerges between the import and export sides. The sales-weighted average total import share declines only slightly from 52.5% to 49.9%, whereas the sales-weighted average total export share falls markedly from 46.0% to 35.7%—a decline comparable to that from shutting down wholesale and retail trade.

²⁰The document is available at https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/06/mapping-the-semiconductor-value-chain_5ba52971/4154cdbf-en.pdf, lastly accessed on May 22, 2026.

The muted response on the import side is notable precisely because our shutdown covers the entire semiconductor supply chain—from raw materials and chemical inputs through foundry equipment and manufacturing machinery to wafers. Even this broad exclusion leaves large Taiwanese firms’ total import exposure intact, reflecting an import basket that spans many products and industries well beyond the semiconductor value chain. The sharp decline on the export side, by contrast, reveals Taiwan’s export reliance on semiconductor final products. The largest firms are directly or indirectly involved in semiconductor exports, as fabricators and assemblers of final chips. Shutting down semiconductor exports not only eliminates the foreign revenue of leading firms like the Taiwan Semiconductor Manufacturing Company (TSMC), but also severs the indirect export exposure of their domestic suppliers, who are themselves large enough to move the sales-weighted average substantially. This is consistent with [Table 5](#), which shows that electronics firms typically have a large number of domestic suppliers: many firms are only a few buyer-seller separations away from international downstream markets, and an export disruption in the semiconductor sector would substantially reduce their exposure to foreign demand.

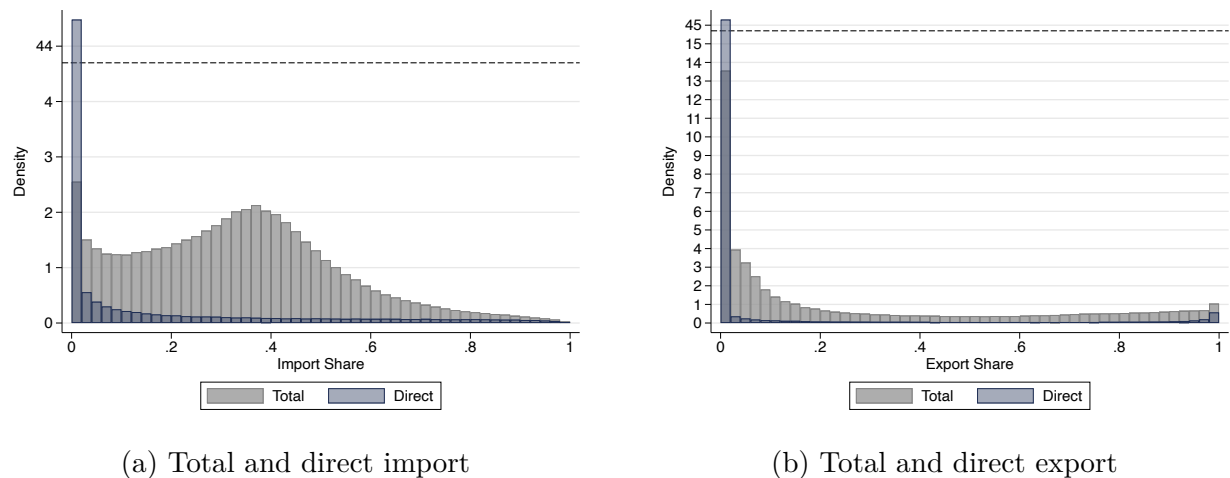


Figure 10: Total Import and Export Excluding Semiconductor-related Products, 2017

Notes: These figures show the total and direct import and export shares after excluding the direct international trade of semiconductor-related products, the relevant HS6 product codes of which are listed in [OECD \(2025a\)](#). The shaded grey histograms depict firm-level total trade shares, while the slate-colored histograms depict direct trade shares.

We summarize the mean import and export shares across all scenarios in [Table 6](#). For each scenario, we report both unweighted means and means weighted by firm-level total sales. The sales-weighted means are substantially higher than the unweighted means, reflecting the outsized trade exposure of larger firms. The effects of a complete disruption of wholesale and retail trade and an exclusion of semiconductor exports are likewise more pronounced under sales-weighted averages.

Additional Results and Robustness. We also consider one additional counterfactual and two robustness checks in Table A.8. In the “Excl. HS 84 85” row, we shut down the direct import and export of machinery (HS chapter 84) and electronics (HS chapter 85), and plot the resulting trade share distributions in Figure A.2. The sales-weighted total import share falls further to 36.1% (from 49.9% in the semiconductor counterfactual) and the sales-weighted total export share to 23.2% (from 35.7%), highlighting that machinery and the broader electronics sector—beyond semiconductor inputs and final goods alone—are critical conduits for Taiwan’s international trade exposure. In the “Alt. domestic sales” row, we apply an alternative measure of domestic sales. In this exercise, we treat the firm’s B2B sales and purchases outside of the cleaned network as domestic contents, instead of excluding them completely as for our baseline sample. It is useful to consider this case so that we can understand how our cleaning process affects indirect trade through domestic networks. The results are mostly consistent with those in the baseline, with total trade shares being slightly smaller as we add an additional term to domestic contents. Since the results are similar, we can be more certain that firms excluded due to having zero total sales or missing labor costs do not play a key role in transmitting international fluctuations. In the “Zero-rated sales as exports” row, we use zero-rated sales in the firm-level VAT filing report as a proxy for firm-level exports, rather than using export volumes from customs records; this yields slightly higher direct and total export shares, both unweighted and weighted. This suggests that export analysis is somewhat sensitive to the choice between customs records and VAT zero-rated sales as the measure of exports.

Table 6: Average Import and Export Shares

	Import Share				Export Share			
	Unweighted		Weighted		Unweighted		Weighted	
	Direct	Total	Direct	Total	Direct	Total	Direct	Total
Baseline	0.039	0.335	0.215	0.525	0.049	0.296	0.243	0.460
Excl. W&R direct trade	0.009	0.138	0.145	0.305	0.017	0.174	0.191	0.340
Excl. semiconductor trade	0.038	0.330	0.198	0.499	0.047	0.277	0.169	0.357

Notes: The table reports average direct and total import and export shares. Unweighted columns report simple firm-level averages. Sales-weighted columns weight firms by total sales. W&R denotes wholesale and retail.

6 Conclusion

This paper documents key stylized facts on Taiwan’s domestic supply chains using newly available firm-to-firm transaction data administered by the Ministry of Finance. Covering

around 1.4 million distinct buyers and sellers, Taiwan’s VAT system reliably captures transactional relationships within the economy. When compared with similar evidence from other countries, our results suggest that certain features of production networks are universally observed: network structures are highly skewed, both across firms and within firms across their trading partners. Among countries with comparable datasets, Taiwan’s empirical patterns offer insights into the structure of small open economies with prominent advanced manufacturing sectors. Our findings also highlight the critical role of the wholesale and retail sectors in facilitating the flow of intermediate goods, both domestically and internationally.

References

- ACEMOGLU, D., V. M. CARVALHO, A. OZDAGLAR, AND A. TAHBAZ-SALEHI (2012): “The network origins of aggregate fluctuations,” Econometrica, 80, 1977–2016.
- ALEKSEEV, M. AND X. LIN (2026): “Export policy for dual-use goods,” Working paper.
- ALFARO-UREÑA, A., M. FUENTES, J. I. GONZÁLEZ, I. MANELICI, AND J. P. VASQUEZ (2026): “The Anatomy of Domestic Production Networks: Evidence from Costa Rica,” Working paper.
- ALFARO-URENA, A., I. MANELICI, AND J. P. VASQUEZ (2022): “The effects of joining multinational supply chains: New evidence from firm-to-firm linkages,” The Quarterly Journal of Economics, 137, 1495–1552.
- AMITI, M., C. DUPREZ, J. KONINGS, AND J. VAN REENEN (2024): “FDI and superstar spillovers: Evidence from firm-to-firm transactions,” Journal of International Economics, 152, 103972.
- ARKOLAKIS, C., F. HUNEEUS, AND Y. MIYAUCHI (2025): “Production Network Formation, Trade, and Welfare,” Working paper.
- AW, B. Y. AND Y. LEE (2008): “Firm heterogeneity and location choice of Taiwanese multinationals,” Journal of International Economics, 75, 167–179.
- AW, B. Y., M. J. ROBERTS, AND D. Y. XU (2011): “R&D investment, exporting, and productivity dynamics,” American Economic Review, 101, 1312–1344.
- BACILIERI, A., A. BORSOS, P. ASTUDILLO-ESTÉVEZ, M. HOEFER, AND F. LAFOND (2026): “Firm-level production networks: What do we (really) know?” Journal of Economic Dynamics and Control, 187, 105313.
- BECKO, J. S. AND D. O’CONNOR (2025): “Strategic (Dis)Integration,” Working paper.
- BERNARD, A. B., E. DHYNE, G. MAGERMAN, K. MANOVA, AND A. MOXNES (2022): “The Origins of Firm Heterogeneity: A Production Network Approach,” Journal of Political Economy, 130, 1765–1804.
- BERNARD, A. B., A. MOXNES, AND Y. U. SAITO (2019): “Production Networks, Geography, and Firm Performance,” Journal of Political Economy, 127, 639–688.
- BORIN, A., F. P. CONTEDECA, F. LEONE, M. MANCINI, AND P. ZOI (2025): “How Global Are Local Value Chains?” Working paper.

- DEMIR, B., A. C. FIELER, D. Y. XU, AND K. K. YANG (2024): “O-Ring Production Networks,” Journal of Political Economy, 132, 200–247.
- DHYNE, E., C. DUPREZ, AND T. KOMATSU (2023): “The Belgian business-to-business transactions dataset 2002-2021,” NBB Working Paper Series, 444.
- DHYNE, E., A. K. KIKKAWA, T. KOMATSU, M. MOGSTAD, AND F. TINTELOT (2025): “Firm Responses and Wage Effects of Foreign Demand Shocks with Fixed Labor Costs and Monopsony,” American Economic Review, 115, 4328–68.
- DHYNE, E., A. K. KIKKAWA, M. MOGSTAD, AND F. TINTELOT (2021): “Trade and Domestic Production Networks,” The Review of Economic Studies, 88, 643–668.
- DHYNE, E., G. MAGERMAN, AND S. RUBINOVA (2015): “The Belgian production network 2002-2012,” NBB Working Paper Series, 288.
- EDMOND, C., V. MIDRIGAN, AND D. Y. XU (2015): “Competition, Markups, and the Gains from International Trade,” American Economic Review, 105, 3183–3221.
- GABAIX, X. (2011): “The granular origins of aggregate fluctuations,” Econometrica, 79, 733–772.
- GROSSMAN, G. M., E. HELPMAN, AND S. J. REDDING (2024a): “When tariffs disrupt global supply chains,” American Economic Review, 114, 988–1029.
- GROSSMAN, G. M., E. HELPMAN, AND A. SABAL (2024b): “Optimal resilience in multitier supply chains,” The Quarterly Journal of Economics, 139, 2377–2425.
- HUNEEUS, F. (2019): “Production Network Dynamics and the Propagation of Shocks,” Working paper.
- HUNEEUS, F., K. KROFT, AND K. LIM (2025): “Earnings Inequality in Production Networks,” Working paper.
- KAO, E. H.-C. AND J.-T. LIU (2022): “Extensive margins of trade and firm survival,” Economics Letters, 218, 110716.
- KAO, E. H.-C., C.-Y. WU, AND J.-T. LIU (2024): “Extensive margins of trade and profitability: Evidence from Taiwan,” Applied Economics Letters, 31, 1569–1573.
- KOMATSU, T. AND E. DHYNE (2023): “Job Ladder over Production Networks,” Working paper.

- LIM, K. (2018): “Endogenous Production Networks and the Business Cycle,” Working paper.
- OECD (2025a): “Mapping the Semiconductor Value Chains: Working Towards Identifying Dependencies and Vulnerabilities,” OECD Science, Technology and Industry Policy Papers.
- (2025b): “Transaction data for evidence-based industrial policy,” OECD Science, Technology and Industry Policy Papers 178, OECD Publishing.
- VARAS, A., R. VARADARAJAN, J. GOODRICH, AND F. YINUG (2021): Strengthening the Global Semiconductor Supply Chain in an Uncertain Era, Boston and Washington, DC: Boston Consulting Group and Semiconductor Industry Association.

A Data Appendix

A.1 Summary Statistics for Full Sample

In our main analysis, we focus on firms with the most complete information that survive the selection process described in Section 3.3. While we report the coverage of our clean sample in Table 3, it is still useful to compare the summary statistics of our baseline to those of the full, raw firm-to-firm transaction data.

Figure A.1 plots the time series of buyer-seller pairs and total transaction volume for the full sample, to be compared with Figure 2 in the main text. On the extensive margin, the full sample contains substantially more buyer-seller pairs throughout the sample period, growing from 55 million to over 100 million (as opposed to from 33 million to 45 million in the cleaned sample), reflecting the large number of small firms excluded by our cleaning process. On the intensive margin, however, the total value of firm-to-firm transactions in the full sample closely tracks that of the cleaned sample, consistent with Table 3, which shows that the cleaned sample captures over 83% of aggregate B2B sales. Together, these patterns suggest that the additional linkages in the full sample are predominantly low-value transactions involving small firms with limited network activity, and that our cleaned sample adequately represents the aggregate transaction volume of Taiwan’s domestic production network.

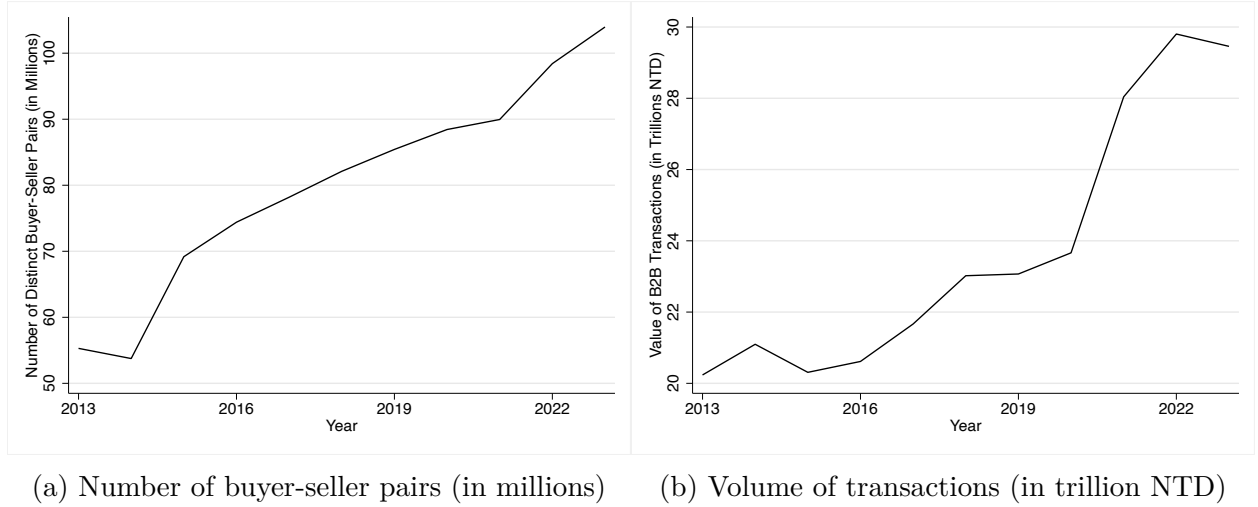


Figure A.1: Development of Taiwanese domestic supply chains, 2013–2022 (Full Sample)

In Table A.1, we can see that once we keep all the small firms that might be missing labor cost information, the average number of linkages decreases from 69.50 to 53.21, and the average network transaction per pair drops from 32.09 million NTD to 14.74 million NTD. In Table A.2, we can see that a lot of firms excluded in our baseline belong to either public sectors (the “other” row) or simply lack the sector information (the “unmatched”

row). These firms now pull down the firm size distribution relative to our baseline. Firms without sectoral information also participate in very few network linkages on average.

Table A.1: Summary statistics for firm-to-firm transactions in 2017, full sample

	Mean	Std.Dev.	Percentiles			
			25th	50th	75th	95th
Link level (N = 78,186,050)						
Transaction amount (in thousands NTD)	277	30869	0.857	2.8	18.11	362.11
Firm level (N = 1,469,417)						
Number of domestic buyers	53.21	704.29	0	1	13	137
Number of domestic suppliers	53.21	143.63	1	15	61	206
Domestic network sales (in millions NTD)	14.74	495.83	0	0.05	2.97	30.61
Domestic input purchases (in millions NTD)	14.74	536.86	0.02	0.031	2.49	27.79

Notes: This table covers the full sample of firms appearing in the B2B transaction data in 2017, prior to the cleaning procedure described in Subsection 3.3. The link-level panel covers all unique domestic buyer–seller pairs; the transaction amount is the total annual value of transactions between each pair, denominated in thousands of New Taiwan Dollars (NTD). The firm-level panel covers all firms appearing as buyers or sellers. Number of domestic buyers (suppliers) is the count of unique downstream customers (upstream suppliers) for each firm in the domestic B2B network. Domestic network sales (input purchases) is the total annual value of a firm’s domestic B2B sales (purchases), denominated in millions of NTD. The mean number of buyers equals the mean number of suppliers, and mean network sales equals mean input purchases, as each domestic transaction is simultaneously recorded as a sale by the seller and a purchase by the buyer.

Table A.2: Firm size distribution of networks by industry

	All firms in 2017 (N = 1,469,417)					
	Share	Firm size			Mean # of links	
	(%)	Mean	p50	p95	Buyers	Suppliers
<i>Economy-wide</i>						
All Industries	100	22.99	0.34	42.89	53.21	53.21
<i>Broad sectors</i>						
Agriculture & mining	0.4	12.58	0.07	47.52	61.07	40.18
Manufacturing	11	85.89	4.09	144.53	52.27	95.80
Utility & construction	9.6	20.63	2.55	44.96	23.21	70.60
Wholesale & retail	42.6	18.80	0.74	46.18	138.36	51.44
Service	20.3	13.73	0.17	29.46	107.47	48.57
Others	3.2	7.54	0.00	9.53	83.82	51.55
Unmatched	12.8	0.11	0.00	0.00	13.14	27.57
<i>Electronics manufacturing</i>						
Electronic parts & components	0.5	560.80	7.03	824.70	47.94	162.69
Computers, electronic & optical	0.3	186.40	3.56	447.91	68.24	158.62
<i>KTI industries</i>						
High-tech manufacturing	0.8	420.34	5.72	676.62	59.87	172.50
Medium-high-tech manuf.	2.3	105.92	5.32	212.98	54.86	125.50
High-tech KI service	1.3	32.95	0.75	53.17	200.95	79.76
KI market service	3.9	13.79	0.80	30.04	43.47	62.89

Notes: The sample consists of all firms in the full B2B network ($N = 1,469,417$) in 2017, prior to the cleaning procedure described in Subsection 3.3. Share of firms reports each sector's fraction of all firms in the full network. Firm size is measured by domestic B2B network sales, denominated in millions of New Taiwan Dollars (NTD). Buyers (Suppliers) reports the mean number of unique domestic downstream customers (upstream suppliers) per firm in the B2B network. Broad sector groupings follow Taiwan's SIC: Agriculture & Mining includes sectors A–B; Manufacturing is sector C; Utility & Construction covers sectors D–F; Wholesale & Retail is sector G; Service covers sectors H–J and L–N, S. The Unmatched category comprises firms that appear in the B2B data but lack a registered industry code. Electronics manufacturing comprises electronic parts & components (two-digit SIC code 26) and computers, electronic & optical products (code 27). KTI denotes knowledge- and technology-intensive industries, classified following Eurostat's definitions; see Table A.6 for details.

A.2 Summary Statistics for 2013 and 2022

Table A.3: Summary statistics for firm-to-firm transactions in 2013

	Mean	Std.Dev.	Percentiles			
			25th	50th	75th	95th
Link level (N = 32,648,384)						
Transaction amount (in thousands NTD)	521.97	52490.67	1.60	7.50	49.46	837.67
Firm level (N = 511,771)						
Number of domestic buyers	63.79	514.36	3	11	35	189
Number of domestic suppliers	63.79	106.25	17	37	74	205
Domestic network sales (in millions NTD)	33.30	916.05	0.66	3.49	12.77	77.60
Domestic input purchases (in millions NTD)	33.30	787.26	0.54	2.54	9.78	75.99

Notes: This table uses the baseline cleaned sample in 2013. The link-level panel covers all unique domestic buyer-seller pairs; the transaction amount is the total annual value of transactions between each pair, denominated in thousands of New Taiwan Dollars (NTD). The firm-level panel covers all firms appearing as buyers or sellers. Number of domestic buyers (suppliers) is the count of unique downstream customers (upstream suppliers) for each firm in the domestic B2B network. Domestic network sales (input purchases) is the total annual value of a firm's domestic B2B sales (purchases), denominated in millions of NTD. The mean number of buyers equals the mean number of suppliers, and mean network sales equals mean input purchases, as each domestic transaction is simultaneously recorded as a sale by the seller and a purchase by the buyer.

Table A.4: Summary statistics for firm-to-firm transactions in 2022

	Mean	Std.Dev.	Percentiles			
			25th	50th	75th	95th
Link level (N = 45,726,024)						
Transaction amount (in thousands NTD)	530.98	62314.22	1.14	4.90	34.66	720.52
Firm level (N = 561,575)						
Number of domestic buyers	73.73	741.31	3	10	35	229
Number of domestic suppliers	73.73	136.80	18	41	85	234
Domestic network sales (in millions NTD)	39.15	936.20	0.58	3.44	13.20	86.30
Domestic input purchases (in millions NTD)	39.15	1381.23	0.54	2.47	9.59	78.60

Notes: This table uses the baseline cleaned sample in 2022. The link-level panel covers all unique domestic buyer-seller pairs; the transaction amount is the total annual value of transactions between each pair, denominated in thousands of New Taiwan Dollars (NTD). The firm-level panel covers all firms appearing as buyers or sellers. Number of domestic buyers (suppliers) is the count of unique downstream customers (upstream suppliers) for each firm in the domestic B2B network. Domestic network sales (input purchases) is the total annual value of a firm's domestic B2B sales (purchases), denominated in millions of NTD. The mean number of buyers equals the mean number of suppliers, and mean network sales equals mean input purchases, as each domestic transaction is simultaneously recorded as a sale by the seller and a purchase by the buyer.

A.3 Sector Codes

The industrial classification system used in Taiwan closely mirrors the Standard Industrial Classification (SIC) framework. In our dataset, industry classifications are obtained from firms' tax registration files, which record a six-digit industry code (BSCD) on an annual basis. Sector codes are defined based on the first two digits of the BSCD code. [Table A.5](#) summarizes the sector codes, corresponding descriptions, and the number of firms included in the sample.

Table A.5: Sector Classification and Sample Firm Counts

Sector	Divisions	Description	Obs. (Full)	Obs. (Baseline)
A	01-03	Agriculture, Forestry, Fishing and Animal Husbandry	5168	1,825
B	05-06	Mining and Quarrying	679	599
C	08-34	Manufacturing	162,637	105,863
D	35	Electricity and Gas Supply	1051	479
E	36-39	Water Supply and Remediation Activities	7144	3,713
F	41-43	Construction	133,279	71,973
G	45-48	Wholesale and Retail Trade	626,078	264,754
H	49-54	Transportation and Storage	31,038	13,000
I	55-56	Accommodation and Food Service Activities	80,258	22,501
J	58-63	Information and Communication	22,643	11,444
K	64-66	Financial and Insurance Activities	28,687	-
L	67-68	Real Estate Activities	41,570	16,280
M	69-76	Professional, Scientific and Technical Activities	49,681	24,985
N	77-82	Support Service Activities	29,196	15,566
P	85	Education	2447	-
Q	86-88	Human Health and Social Work Activities	804	-
R	90-93	Arts, Entertainment and Recreation	15,658	-
S	94-96	Other Service Activities	43,846	8,584

Notes: Division refers to the first two digits of the industrial code. Sector O (Public Administration and Defence; Compulsory Social Security) is excluded from the table due to confidentiality concerns.

Additionally, we define broader sectoral groupings for analysis. Specifically, Agriculture and Mining include sectors A and B; Manufacturing corresponds to sector C; Utility and Construction encompass sectors D, E, and F; Wholesale and Retail refer to sector G; Service covers sectors H, I, J, L, M, N, and S; and Other comprises sectors K, O, P, Q, R, and X. Knowledge- and technology-intensive industries are defined according to Eurostat's definitions, and are described in detail in [Table A.6](#).

Table A.6: Definition of High-Tech Industries

Division Code	Description
High-Technology Manufacturing	
20	Manufacture of Pharmaceuticals and Medicinal Chemical Products
26	Manufacture of Electronic Parts and Components
27	Manufacture of Computers, Electronic and Optical Products
Medium-High-Technology Manufacturing	
18	Manufacture of Chemical Material and Fertilizers
19	Manufacture of Other Chemical Products
28	Manufacture of Electrical Equipment
29	Manufacture of Machinery and Equipment
30	Manufacture of Motor Vehicles and Parts
31	Manufacture of Other Transport Equipment and Parts
High-Tech Knowledge-Intensive Services	
59	Motion Picture, Video and Television Programme Production, Sound Recording and Music Publishing Activities
60	Programming and Broadcasting Activities
61	Telecommunications
62	Computer Programming, Consultancy and Related Activities
63	Information Service Activities
72	Scientific Research and Development
Knowledge-Intensive Market Services	
50	Water Transportation
51	Air Transport
69	Legal and Accounting Activities
70	Activities of Head Offices; Management Consultancy Activities
71	Architecture and Engineering Activities; Technical Testing and Analysis
73	Advertising and Market Research
74	Specialized Design Activities
75	Veterinary Activities
76	Other Professional, Scientific and Technical Activities
78	Employment Activities
80	Security and Investigation Activities

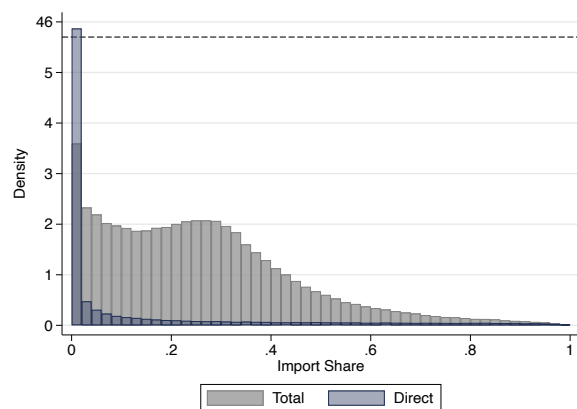
Note: High-tech industries are classified according to Eurostat's NACE Rev. 2 definitions, with corresponding divisions aligned to Taiwan's SIC 9th revision.

A.4 Total Trade Share: Product Codes and Additional Results

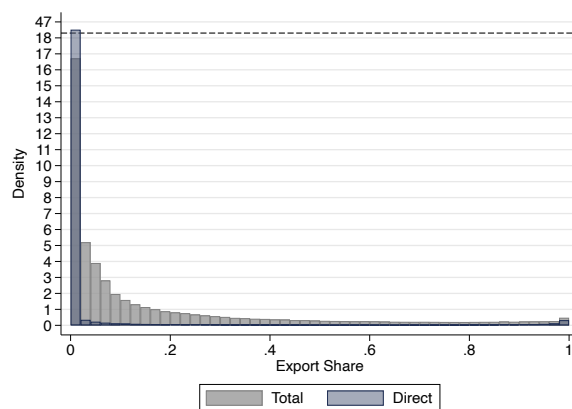
Table A.7: Semiconductor Product Codes (HS Classification)

Category	HS Codes
<i>Input</i>	
Inputs for Wafer Fabrication	370130, 370199, 370790
Silicon Wafers	381800
Foundry Inputs	900120, 900190, 900219, 900220, 900290, 901210, 901290, 903141
Manufacturing Equipment	841459, 841950, 842129, 842139, 842199, 848610, 848620, 848630, 848640, 848690, 900699, 903082, 903084, 903300
Photosensitive Semiconductor Devices	854140, 854150
<i>Output</i>	
Final Products (Chips)	854110, 854121, 854129, 854130, 854160, 854190, 854231, 854232, 854233, 854239, 854290, 852351, 852352, 852359, 853290, 853390

Notes: HS codes based on the OECD (2025) “Mapping the Semiconductor Value Chain: Working towards identifying dependencies and vulnerabilities.” The classifications are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.



(a) Total and direct import



(b) Total and direct export

Figure A.2: Total Import and Export Excluding Machinery (HS 84) and Electronics (HS 85), 2017

Notes: These figures show the total and direct import and export shares after excluding the direct international trade of machinery (HS chapter 84) and electronics (HS chapter 85). The shaded grey histograms depict firm-level total trade shares, while the slate-colored histograms depict direct trade shares.

Table A.8: Robustness Checks of Average Import and Export Shares, Unweighted and Weighted by Total Sales

	Import Share				Export Share			
	Unweighted		Weighted		Unweighted		Weighted	
	Direct	Total	Direct	Total	Direct	Total	Direct	Total
Excl. HS 84 85	0.027	0.258	0.129	0.361	0.032	0.173	0.108	0.232
Alt. domestic sales	0.039	0.270	0.215	0.477	0.049	0.280	0.243	0.448
Zero-rated sales as exports					0.059	0.333	0.282	0.536

Notes: The table reports robustness checks on the average direct and total import and export shares from [Table 6](#). “Excl. HS 84 85” excludes all direct trade in machinery (HS chapter 84) and electronics (HS chapter 85). “Alt. domestic sales” treats each firm’s B2B sales and purchases outside the cleaned network as domestic content, rather than dropping them entirely as in the baseline. “Zero-rated sales as exports” uses VAT zero-rated sales as a proxy for firm-level exports instead of customs records; import shares are unchanged and therefore not reported.